

HLJBJ 混凝土自动上料搅拌车使用说明书

Operation Manual of HLJBJ Automatic Charging Concrete Mixer Truck

非常感谢您选购本公司的自动上料搅拌车，请您在使用设备前，仔细阅读使用说明书。本说明书将向您提供设备的性能、技术参数、使用、维护及注意事项的说明。用户必须严格按照本说明书所述方法和各项规定，安全使用自动上料搅拌车并正确地进行操作和维护保养。正确的操作日常维护保养是实现安全生产、减少故障，延长使用寿命的基本保证。自动上料搅拌车的操作者、维护人员及技术人员应在仔细地阅读本说明书的基础上，使自己对自动上料搅拌车相关原理和使用、操作要领有较透彻的了解。

Thank you for selecting our automatic charging mixer truck. Please read this Manual carefully before use. In this Manual, you will find instructions on performance, technical parameters, usage, maintenance and considerations of the equipment. Do strictly follow the methods and requirements stated in this Manual to safely use the automatic charging concrete mixer truck through correct operation, service and maintenance. Correct operation and daily maintenance are the basic guarantee for safe production, failure reduction and longer service life. Operators, maintainers and technicians of the automatic charging mixer truck will have a thorough understanding on principles and operation rules of the automatic charging mixer truck after carefully reading the Manual.

《使用说明书》被认为是车辆设备的一部分，应与车辆一起保存，以备随时查阅。

The Operation Manual is deemed as part of the vehicle equipment and shall be kept therewith for your easy reference.

公司对产品将会不断改进和完善，其变化恕不能一一通知您，敬请原谅。

We will constantly improve and perfect the product. Please forgive us for the inability to inform you of all changes.

感谢您对本公司的信赖与支持！

Thank you for your faith and support!

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第一章 用途、性能及注意事项

Chapter I Use, Performance and Considerations

LZ3500 混凝土自动上料搅拌车已注册国家五项专利，广泛应用于建筑行业。能大大提高工作效率。该产品装有液压电子称，在上料时对石子、沙子能够精确计量，自动定量上水，使混凝土配比更标准。该产品是结合现有混凝土运输车和搅拌站功能为一体的设备。可自动上料、计量、搅拌到卸料，实现运输和现场施工作业。

LZ3500 Automatic Charging Concrete Mixer Trucker has registered five national patents and is widely used in the construction industry. It can greatly improve working efficiency. The built-in electronic hydraulic scale allows the equipment to accurately measure pebbles and sand and automatically feed quantified water in the charging process so that the concrete mix ratio is more accurate. This equipment is an integration of a concrete carrier vehicle and a mixing plant. It can realize auto charging, measuring, mixing and discharging, capable of both transport and on-site construction.

混凝土自动上料搅拌车具有设计合理、结构紧凑、操纵灵活、维修方便等优点。

The automatic charging concrete mixer truck has such advantages as rational design, compact structure, flexible operation and easy maintenance.

注意事项

Precautions

①严禁超载！超载将导致不可预见的安全事故的发生。

①Overload prohibited! Overload may lead to unpredictable accidents.

②驾驶员及有关人员在使用搅拌车之前，必须认真仔细地阅读使用维护说明书或操作维护保养手册，按规定的事项去做。否则会带来严重后果和不必要的损失。

②Driver and relevant personnel shall carefully read the Maintenance Manual or the Operation and Maintenance Manual before use and strictly follow the instructions during usage. Or serious consequence and unnecessary losses may occur.

③驾驶员穿戴应符合安全要求，并穿戴必要的防护设施。

③The wear of driver shall conform to safety requirement, plus protective clothing required.

④在作业区域范围较小或危险区域，必须在其范围内或危险点设置警告标志。

④Warning signs shall be set within or at dangerous points in a narrow operation area or a hazardous area.

⑤绝对严禁驾驶员酒后或过度疲劳驾驶作业。

⑤Drunk driving or over-fatigue driving is absolutely forbidden.

⑥进入搅拌罐或进行检查之前，必须在驾驶方向盘上贴警示标签并断开电池电源。

⑥Stick the warning sign to the steering wheel and disconnect battery supply before entering or checking the mixing drum.

⑦要在搅拌车停稳之后，在有蹬梯扶手的地方上下搅拌车。切勿在搅拌车作业或行走时跳上跳下。

⑦Get on or off the mixer truck where a staircase handrail is equipped after the truck completely stops. Jumping on and off the mixer truck when it is in operation or movement is forbidden.

⑧维修搅拌车需要举臂时，必须把举起的动臂垫牢，在维修没完成情况下，动臂绝对不会落下。

⑧Cushion the lifting arm firmly when arm lifting is required in maintenance to ensure this arm will not fall before the maintenance finishes.

⑨在搅拌车行驶前，收起延伸料槽并用挂钩固定牢靠。卸料槽固定在行驶位置并锁紧手柄。

⑨Put down the extended chute and fix it firmly to the hook before the mixer truck moves. The discharge chute is fixed to the driving position and locked to the handle.

⑩行驶时应时刻牢记搅拌车重心高且装满混凝土的搅拌罐旋转时向右偏心。车辆向左转弯时应比向右转时更当心,在运输混凝土时严禁搅拌罐转速超过 10 转/分钟。

⑩Keep in mind during the driving that the mixer truck has a high center of gravity, which shifts to the right when the mixing drum rotates with a full load of concrete. When a turning is performed, be more careful if it is a left turn. The maximum rotation speed of the mixing drum loaded with concrete is 10 rpm.

⑪严禁靠近旋转的搅拌罐运动部件,严禁手接触旋转的滚轮及传动轴。

⑪Keep off the rotating parts of the mixing drum. Keep hands off the rotating roller and transmission shaft.

第二章 整机主要性能和参数

Chapter II Performance and Parameters

发动机型号: 玉柴 4105 增压发动机

Engine model: YuChai 4105Turbocharged engine

变 速 箱: JBJ-320 分体变矩器

Gearbox: JBJ-320 Split Torque Converter

桥: 全液压驱动, 国产车桥

Axle: Fully-hydraulic driven, 2-wheel steering, made in China

刹车形式: 气刹制动器, 双独立回路

Brake method: Air brake system with dual independent circuits.

驱动形式: 四驱

Driving mode: 4-wheel drive

额定功率: 85kw

Rated power: 85 kw

额定转速: 2400r/min

Rated rotation speed: 2,400 r/min

最高时速: $\leq 35\text{km}$

Max. speed per hour: $\leq 35\text{ km}$

最小转弯半径: 4.2m

Minimum turning radius: 4.2 m

搅拌形式: 液压马达驱动

Mixing mode: Hydraulic motor driven

罐体容积: 5.5m^3

Tank capacity: 5.5 m^3

罐体出料: 3.5m^3

Tank discharge: 3.5m^3

搅拌速度: 20r/min

Mixing speed: 20r/min

斗 容 量: 0.65m^3

Dipper capacity: 0.65 m^3

轮胎型号: 12R22.5

Tire model: 12R22.5

称 重: 电子液压称

Weigh: Electronic hydraulic scale

供水形式: 自带供水系统(24V 电控自动加水系统)

Water supply method: Built-in water supply system (24V electronic-control auto water feeding system)

水箱容量: 850L

Water tank capacity: 850 L

液压油箱容积: 120L

Oil tank capacity: 120L

外形尺寸: 7600X2872X3374mm

External dimensions: 7600 × 2872 × 3474 mm

整车质量: 8300kg

Total weight: 8300 kg

第三章 搅拌车主要构成及功能

Chapter III Main Components and Functions of the Mixer Truck

混凝土自动上料搅拌车是一种适用于小型工程建设,集自动上料、搅拌、运输、自动卸料于一体的混凝土搅拌运输车。采用大功率柴油发动机,液力变矩器,保证行走与搅拌同时工作所需动力。主要结构由:行走底盘、发动机、液力变矩器、前后四驱桥、上料装置、卸料斗、加长卸料斗、机架、搅拌罐、搅拌专用行星减速机、液压系统、电控系统、自动上水系统等组成,其特征如下:

The automatic charging concrete mixer truck is applicable to small-scale construction projects, which integrates the automatic charging, mixing, transport and automatic discharging into a whole. The high-power diesel engine and hydraulic torque converter are used, ensuring the power required during moving and mixing. The mixer truck consists mainly of: chassis, engine, hydraulic torque converter, front and rear 4-wheel drive axle, charging device, discharging hopper, extended discharging hopper, frame, mixing drum, special planetary reducer for mixing, hydraulic system, electric control system and automatic water supply system, which features as follows:

1.行走系统由发动机通过液力变矩器、前后传动轴到前后桥。采用液压转向四驱形式,大大减小转弯半径,更适于工地上工作,特别是狭小空间施工。采用液力变矩器减少对发动机的冲击、延长了发动机的寿命。行走采用无极变速电控阀,方便操控。

1. The driving system operates by engine driving the front and rear axle through hydraulic torque converter and front and rear transmission shaft. The hydraulic steering and 4-wheel drive is used, greatly reducing the turning radius, which enables the mixer truck to be more suitable for construction site, especially for the construction in limited space. The hydraulic torque converter reduces the impact to the engine, extending the service life of engine. The continuously variable transmission (CVT) electric control valve is applied to the drive, easy to operate and control.

2.液压系统采用恒功率变量柱塞泵,减少发动机的功率损耗,。

2. The constant-power variable displacement position pump is used for the hydraulic system, reducing the power loss of engine.

3.搅拌罐的旋转由液压马达通过专用行星减速机带动,行星减速机的输出轴可以实现一定角度的摆动,以满足恶劣路况下搅拌罐的跳动。搅拌车的搅拌罐加有控速阀,能够精确控制搅拌罐搅拌转速。

3. The rotation of mixing drum is actuated by the hydraulic motor through the special planetary reducer whose output shaft can swing within a certain angle, meeting the jump requirements of mixing drum in poor road conditions. The mixing drum of mixer truck is equipped with speed-controlling valve, which can control the mixing speed precisely.

4. 干料上料，按照设计好的配比由铲臂、举升油缸、铲斗铲料后，电子称称重，自动上料。用于出料的卸料槽可以旋转并通过锁紧机构固定在适当的位置。用于加长卸料槽长度的延伸料槽可折叠并放置于其固定位置，并用挂钩固定在卸料槽上。

4. Charge of dry materials: the materials with designed proportion will be charged automatically after being spaded through bucket arm, lifting cylinder and bucket and weighed by electronic scale. The discharge chute can rotate and be fixed in the proper position through locking device, while the extended chute for extended discharge is folding, placed in the fixed position, and fixed on the discharge chute.

5. 进水系统带有定量自吸泵，由定量控制器设定实现自动定量进水。

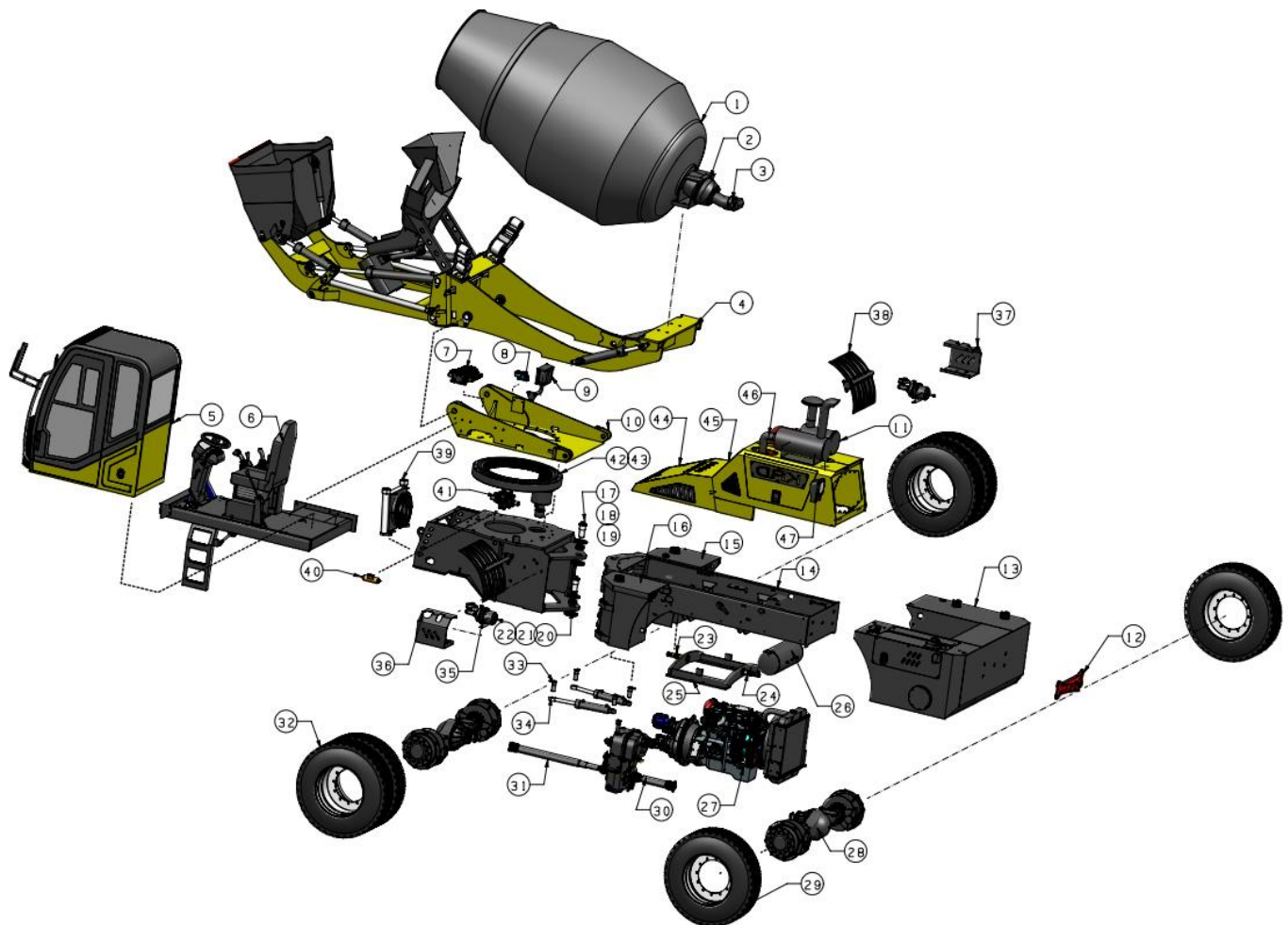
5. The water system is equipped with the quantitative self-priming pump, realizing the automatic quantitative water inlet by setting the quantitative controller.

6. 搅拌罐通过三点固定和支撑在底架上，前端与固定在底架上的齿轮箱输出法兰联接，该法兰可以改变倾斜角度。搅拌罐通过焊装的轨道支撑在内有锥型轴承的 2 个滚轮上。进料、搅拌和出料由压制成型的叶片系统来完成。搅拌罐正向旋转为进料、混凝土搅拌；反向旋转，在螺旋叶片推动下、混凝土由出料口出料。

6. The mixing drum is fixed and supported on the chassis by three-point, and its front end links the output flange of gearbox fixed on the chassis, which can change the angle of inclination. The mixing drum is supported on 2 rollers with cone bearing through the welded track. The charge, mixing and discharge are realized through the blade system molded by pressing. When the mixing drum is rotating forward, the materials are being charged and mixed, while it is rotating reversely, the materials are being discharged through the discharge door driven by the helical blade.

示意图：铲料位置图 1、上料位置图 2

Schematic diagram: Location Plan 1 for Spading and Location Plan 2 for Charge



零件名称	Part Name
1.罐车罐	1.Tank truck tank
2. 转罐减速机	2.Transfer tank reducer
3. 转罐马达	3.Transfer tank motor
4.罐架及上料工作装置总成	4.Tank rack and feeding device assembly
5.驾驶室	5.Cab
6.底板总成	6.Base plate assembly
7.分配器 DL15.6-4	7.Distributor DL15.6-4
8.优先阀	8.Priority valve
9.灯架	9.Lighting frame.
10.罐车上盘	10.Pan above the tank truck
11.空滤总成	11.Air filter assembly
12.牵引	12.Tow
13.水箱配重总成	13.Water tank balance assembly
14.罐车后车架	14.Rear frame
15.燃油箱总成	15.Fuel tank assembly
16.液压油箱总成	16.Hydraulic tank assembly
17.镀锌销轴/交接无帽	17.Galvanized pin shaft/transfer without cap
18.上压盖	18.Top pressure cover

19.关节轴承 GE60	19.Joint bearing GE60
20.镀锌销轴/交接带帽	20.Galvanized pin shaft/transfer coupling cap
21.下压盖	21.Lowering cover
22.铁套 70X60X18	22.Iron casing 70x60x18
23.镀锌销轴/摇摆带帽	23.Galvanized pin shaft/swing cap
24.铜套 50X40X45	24.Copper sleeve 50x40x45
25.罐车摇摆	25.The tank truck is swaying
26.气筒	26.Inflator
27.动力传的系统总成	27.Power transmission system assembly
28.驱动桥	28.Driven axle
29.轮胎总成	29.Tire assembly
30.传动轴	30.Transmission shaft
31.传动轴	31.Transmission shaft
32.轮胎总成	32.Tire assembly
33.转向销轴	33.Steering pin shaft
34.转向油缸	34.Steering cylinder
35.断气刹总成	35.Air brake assembly
36.罐车断气刹盒	36.Container truck air brake box
37.罐车断气刹盒	37.Container truck air brake box
38.挡泥瓦	38.Fender
39.散热器	39.Radiator
40.高压过滤器	40.High-pressure filter
41.分配器	41.Dispenser
42.回转支撑	42.Rotary support
43.回转减速机	43.Rotary reducer
44.机罩	44.Tank car hood
45.--	45.--
46.消音器	46.Silencer
47.电子临工大灯	47.Electronic temporary worker headlights

注意加注黄油！

Do fill grease!

第四章 搅拌车操作方法

Chapter IV Operation Method of Mixer Truck

在操作之前，操作者必须仔细阅读手册，尤其是安全细节和所有的提醒、警告，并要熟悉工作中的常用手势。

The operator must carefully read the manual, especially the safety details and all cautions, warnings before operation and be familiar with common hand signals.

搅拌机操作的重要步骤：

Important procedures for operation of mixer:

工作前先检查水箱是否有水，没有需要先用水泵抽水加满；查看柴油箱的油位，是否能保证使用；检查液压油箱油位；查看轮胎是否损伤、气压表正常、螺栓是否松动等。确认上水阀在开启状态，洗车水阀在关闭状态，洗车水桶放气开关关闭状态。在各润滑部位加注润滑油。

Before operation, check the water tank and fill it with water through pump in case of no water; check the oil level of diesel tank for normal operation; check the oil level of hydraulic oil tank; check whether the tire is free from damage, barometer is normal and the bolt is loosened. In addition, check that the water inlet valve is open, truck wash water valve is close and the air relief switch of truck wash bucket is close. And add lubricating oil at each location to be lubricated.

发动车前在启动或关闭发动机前，确保各控制手柄处于中间位置，确认后发动搅拌车。

Before the engine is started or shut down, ensure each control handle in middle position, then start the mixer truck.

发动后检查工作台仪表是否正常，当气压表达到 4mpa 时，其他仪表正常后，可踩下离合踏板，挂低档，松开断气刹，搅拌车开始行走。

After it is started, check whether the instruments on the operator's platform are normal or not; when the barometer reaches 4 mpa and other instruments get to normal state, the mixer truck will begin moving by pressing the clutch pedal, shifting into low gear and releasing the air brake.

首先把搅拌罐控制杆推至开启位置，打开搅拌罐的搅拌模式；然后打开上水定量控制器上水；打开电子称控制器，开始上料，推动铲斗举升大臂控制手柄将大臂放到最低，斗子放平铲料、收斗、举升臂、称重，称重后继续上升，铲斗仓口对准罐口后，开启仓门卸料，卸料后关闭仓门，进行下次上料。当称重累计超出要求重量时，超出部分打开仓门卸掉，然后再上到罐口卸料。直到水泥、沙石达到规定重量。加料过程中搅拌罐一直慢速转动，加好料后，提高搅拌速度，待搅拌均匀后，到需要位置卸料。将加长卸料斗接到卸料斗上，搅拌罐控制杆扳到卸料位置（搅拌罐反转），卸料完毕，收好加长卸料斗。重复操作继续工作。

First push the joystick of mixing drum to the open position to start the mixing; then feed water by opening the quantitative controller of water; open the electric scale controller, then the charge starts; drive the bucket to lift the control handle of arm and lower the arm to the lowest position, then the bucket lays flat and spades the materials, retracts, lifts arm and weighs, then it continues to rise; when the door of bucket aligns with the mouth of drum, the bucket will open the door and discharge, then close the door after discharge, and so on. When the cumulative weight weighed exceeds the specified weight, the excess will be unloaded through the door, and then the rest will be lifted to the mouth of drum. The above process will continue until the cement, sand and stone reach the specified weight. In the process of charge, the mixing drum keeps rotating in low speed and increases its mixing speed after the charge is completed. When the materials are well mixed, the discharge can be carried out in the required position. Connect the extended discharge hopper to the discharge hopper and pull the joystick of mixing drum to the discharge position (the mixing drum rotates reversely). Put the extended discharge hopper away when the discharge is over. Keep the mixer truck operating by repeating the procedures above.

每日工作完成后需要清洗搅拌罐，以防水泥干结，造成罐内容量减小。操作步骤：打开上水阀，加水进罐后，空转数分钟，然后高速将水卸出。彻底清洗车外部并将加长卸料斗固定好。停好搅拌车，拉手刹、关电子称、熄火。

The mixing drum needs to be cleaned after daily work to prevent the cement from dry and hard, reducing the drum capacity. Operating procedures: After water inlet valve is opened and the tank is filled with water, keep the mixing drum idle for several minutes and then discharge the water at high speed. Wash the exterior thoroughly and secure the extended hopper in place. Pull handbrake, close electronic scale and shut down the engine to stop the mixer truck.

第五章 搅拌车的保养与维修

Chapter V Maintenance and Repair of Mixer Truck

在搅拌机设计中已充分考虑了操作人员及其他工作人员的安全。但安全仍依赖于正确的工作、定期的检查和及时的维护保养。

The safety of operators and other staff has been taken into full account in the design of mixer. However, the proper operation, regular check and timely maintenance are the key to safety.

日常准备工作

Daily Preparations

搅拌机启动前检查如下项目：

Do the following checks before starting the mixer:

每日查看搅拌机，检查是否有齿轮油泄露、软管腐蚀、液压油泄露、滚轮或卸装置破坏或故障等问题。检查卸料槽是否正确固定。检查液压油箱中液压油油位，不足时加液压油；检查齿轮箱中齿轮油油位，不足时加齿轮油；滚轮—表面涂抹黄油；滚轮轴承—加注黄油；上料、卸料机构—加注黄油；调整轨道保护带并清理轨道；水箱—加满水并确保处于卸压状态；在混凝土容易粘附的搅拌机外表上喷涂一层柴油和水蜡混合液或近似溶液，这样有利于防止混凝土粘附，便于清洗。每日操作结束后应清洗搅拌机。

Examine the mixer daily and check whether the problems including leakage of gear oil, corrosion of hose, leakage of hydraulic oil, damage or fault of roller or discharging device exist. Check whether the discharge chute is secured properly. Check the hydraulic oil level in the hydraulic oil tank and fill if not sufficient; check the gear oil level in the gearbox and fill if not sufficient; roller - surface is greased; roller bearing - fill grease; charge and discharge mechanism - fill grease; adjust the track guard belt and clear the track; water tank - fill it with water and ensure it in pressure relief state; spray a layer of diesel and water-wax mixture or similar solution on the exterior surface of mixer where the concrete is easy to adhere to, to prevent the adhesion of concrete and be sure it's easy to clean. The mixer should be cleaned after daily work.

搅拌机停放一夜，即使气候较暖，油温也会降低，因此在每天使用前必须先预暖以下，直至液压油油温上升。暖气候发动机怠速预热 3-5 分钟；寒冷气候，发动机怠速预热 10-15 分钟。

The oil temperature will decrease even in a warm climate after the mixer stops overnight, so the mixer must preheated until the temperature of hydraulic oil increases before daily use. Preheat the engine for 3 to 5 minutes in idle speed in warm climate, while preheat the engine for 10 to 15 minutes in idle speed in cold climate.

定期维护保养，按期更换液压油与齿轮油。

Regularly maintain and periodically replace the hydraulic oil and gear oil.

保养周期

Maintenance Period

注意：首期使用达 50-100 小时时更换滤清器芯，否则液压系统将不予以质量担保。

Notes: replace the filter core when it is used for 50 to 100 hours at the initial stage, or the hydraulic system will not be warranted.

A、首期保养： 初期使用 50-100 小时后

A. Maintenance at the initial stage: after 50 to 100 hours after initial use

A1、排放干净液压油箱中原有脏液压油，并用新油冲洗一遍，再用新液压油加满。

A1. Empty the original dirty hydraulic oil in the hydraulic oil tank and wash the tank with new oil, then fill the tank with new hydraulic oil.

A2、更换滤清器滤芯

A2. Replace the filter element

A3、检查并加紧以下螺栓

A3. Check and tighten the following bolts

-- 齿轮箱固定螺栓

-- Fixed bolt of gearbox

-- 齿轮箱与罐体连接螺栓

-- Connecting bolt between gearbox and drum

-- 罐体与大架固定螺栓

-- Fixed bolt of drum and large frame

-- 传动轴连接螺栓

-- Connecting bolt of transmission shaft

-- 泵固定螺栓

-- Fixed bolt of pump

-- 水罐底座固定螺栓

-- Fixed bolt of water tank base

日检

Daily Check

检查液压油是否清洁和油位是否超过油位计的中间位置，不足时加油。注意：当油温底时不宜加太满。（高配的每 4 小时定量加注一次黄油）

Check whether the hydraulic oil is clean and the oil level exceeds the middle position of oil level gauge and fill oil if not sufficient. Notes: do not overfill when the oil temperature is low. (For the higher configuration, fill grease once quantitatively every 4 hours)

50 小时或周检

50-hour or Weekly Check

检查液压油箱中液压油油位，确保超过油位计中位，不足时加油；给滚轮和轨道加注黄油。

Check the oil level in the hydraulic oil tank and ensure it over the middle position of oil level gauge, and fill oil if not sufficient; add grease on roller and track.

200 小时或月检

200-hour or Monthly Check

检查 50 小时的保养事项；

Check and maintain the items after 50 hours;

用低压黄油枪给传动轴加黄油，给万向节加黄油直至油从四周渗出，给传动花键上黄油，直至渗出。

Add grease on the transmission shaft with low-pressure grease gun and add the grease to the cardan joint and transmission spline until the grease is oozed.

检查并加紧以下螺栓

Check and tighten the following bolts

- 齿轮箱固定螺栓
- Fixed bolt of gearbox
- 齿轮箱与罐体连接螺栓
- Connecting bolt between gearbox and drum
- 罐体与大架固定螺栓
- Fixed bolt of drum and large frame
- 传动轴连接螺栓
- Connecting bolt of transmission shaft
- 泵固定螺栓
- Fixed bolt of pump
- 水罐底座固定螺栓
- Fixed bolt of water tank base

500 小时或季检

500-hour or Quarterly Check

检查保养 200 小时的项目

Check and maintain the items after 200 hours

更换滤芯。

Replace the filter element.

更换液压油，每操作 500 小时或 6 个月后，排放、冲洗并加满液压油箱。

Replace the hydraulic oil: discharge, wash and fill the hydraulic oil tank with hydraulic oil every 500 hours of operation or after 6 months.

搅拌车的检查与调整

Check and Adjustment of Mixer Truck

本搅拌车在出厂前各部位均已调整至最佳状态，客户在使用新机时无需调整。待工作一段时间后一些部位可作适当的调整，具体调整方法如下：

Each component of the mixer truck has been adjusted to the best state before the mixer truck is delivered out of factory, so it is unnecessary for the client to adjust the new mixer truck. After operation for a period of time, some components may be adjusted properly with the following methods:

A、发动机：详见柴油机说明书

A. Engine: refer to the Instructions of Diesel Engine for further details

B、离合器：

B. Clutch:

1. 离合器踏板自由行程的调整

1. Adjust the free travel of clutch pedal

为保证离合器在传递动力时不打滑，离合器分离轴承与分离杠杆之间应保持一定间隙，具体反应到踏板上应有 25-30 毫米的自由行程距离。使用过程中，由于磨擦片磨损，分离杠杆与分离轴承之间的间隙将逐渐减少，甚至消失。因此必须定期检查和调整踏板的自由行程。

In order to ensure nonskid of clutch during transmission of power, certain clearance should be kept between the release bearing and release lever of clutch, which, specifically, should enable the pedal to have free travel distance of 25 to 30 mm. In the process of operation, the clearance between the release lever and release bearing will decrease gradually; even disappear as the friction plate is

being worn. Therefore, the free travel of pedal should be checked and adjusted regularly.

自由行程调整是改变离合器拉杆的长度来实现的, 放长离合器拉杆, 踏板自由行程便减少, 缩短离合器拉杆, 自由行程增大。

The adjustment of free travel is realized by change of the clutch lever. The free travel of pedal decreases as the clutch lever lengthens, and vice versa.

2. 离合器分离杠杆位置的调整

2. Adjust the position of release lever of clutch

离合器的 2 个分离杠杆端面应在同一平面内, 彼此相差不超过 0.15 毫米。用转动调节螺栓的方法进行调整。

Ends of 2 release levers of clutch should be in the same surface and the spacing between them should not exceed 0.15 mm. The adjustment is realized by turning of the adjustment bolt.

C、传动轴:

C. Transmission shaft:

搅拌车有六个万向节, 万向节十字轴上有注油嘴, 每工作 150 小时应注润滑油脂。

The mixer truck is equipped with six cardan joints and the cross axle of cardan joint is provided with oil injection nozzle. The lubricating grease should be filled every 150 hours of operation.

D、驱动桥:

D. Drive axle:

主、被动齿轮已按接触区标准进行选配, 因此, 可长期使用不应进行调整, 只有在更换损坏零件或因零件磨损而移动时, 方需进行调整。调整时应注意以下几项:

The driving and driven gears have been assembled based on the standard of contact zone, so they can be used for long period without adjustment and will be adjusted only when the damaged parts are replaced or they are moved resulting from the wear of parts. The advices below should be followed during adjustment:

(1) 主动齿轮的轴承间隙: 由主动齿轮上的轴承内圈背后的垫座来调整, 调整后应该没有间隙而又能自由转动。

(1) Bearing clearance between driving gears: the adjustment is realized through the pillow at the back of bearing inner ring on the driving gear. After the adjustment is completed, no clearance exists and the driving gear can rotate freely.

(2) 主、被动齿轮的齿侧间隙: 在主动齿轮的法兰盘半径 45mm 的周围上测时, 其位移(弧长)应在 0.2-0.4mm 之间。

(2) Gear backlash between driving gear and driven gear: on the upper side around the radius of 45 mm of flange on the driving gear, the displacement (arc length) should be within 0.2 - 0.4 mm.

(3) 每行驶 100 小时后, 检查驱动桥油面, 正确的油面高度为不低于加油口下缘 10mm, 否则要添加。每行驶 300 小时后, 应更换齿轮油, 本机应加 18 号双曲线齿轮油。

(3) Check the oil level of drive axle for travel of 100 hours. The correct oil level height shall not lower than 10 mm under the oil filler, and, if not, the oil needs to be filled. The gear oil should be replaced for travel of 300 hours. The 18# hypoid gear oil should be filled for this drive axle.

(4) 每行驶 100 小时后, 应更换一次轮壳轴承润滑脂, 调整轴承预紧度到合适为止。

(4) The lubricating grease of wheel bearing should be replaced for 100 hours of running and the pretension of bearing should be adjusted properly.

E、制动系统

E. Braking system

本机制动采用液压制动系统，制动踏板全行程 160mm，自由行程为 8-10mm，当发现制动踏板太低时应检查：

The hydraulic braking system is used for braking provided with brake pedal with total travel of 160 mm and free travel of 8 - 10 mm. When the brake pedal is too low, do the following checks:

(1) 管道是否漏油。

(1) Check the pipe for oil leakage.

(2) 蹄片与鼓的间隙是否过大。

(2) Check whether the clearance between the brake shoe and drum is too large.

(3) 管路中是否有空气。

(3) Check the line for air existence.

每工作 200 小时，应检查制动蹄片的磨损。当蹄片磨损到使踏板行程过大时，必须调整蹄片和鼓之间的间隙，其过程如下：

Check the wear of brake shoe for operation of 200 hours. When the travel of pedal is too large due to the wear of brake shoe, the clearance between the brake shoe and drum should be adjusted with procedures shown below:

1、顶起需调整的车轮。

1. Jack up the wheel to be adjusted.

2、在制动底板的背面找到偏心轮螺栓，转动偏心轮螺栓，直到蹄片与鼓接触，然后放松偏心轮螺栓，直到车轮自由转动并且蹄片与鼓不再磨擦时为止。

2. Find the eccentric bolt at the back of brake bottom plate and turn it until the brake shoe contacts the drum, then release the eccentric bolt until the wheel moves freely and the no friction exists between the brake shoe and drum.

在更换制动液以及贮油壶干底后添加新制动液时，需要将制动管路及泵中空气放净。其方法如下：

When the brake fluid is replaced and the new brake fluid is added after the oilcan is used up, all air in the brake line and pump is required to be exhausted with the method shown below:

1、将制动踏板连续地踏下数次，松开放气阀，放出管内空气，旋紧气阀，松开制动踏板。如此重复数次，直到油路内空气排完为止。

1. Step on the brake pedal continuously for several times and release the air valve to exhaust the air, then tighten the air valve and release the brake pedal. Repeat in the above steps for several times until all air in the oil line is exhausted.

2、放气过程中要随着贮油壶油面降低而不断添加制动液,不能让贮油壶干底，全部放气完毕后，应把制动液加足。

2. During exhaustion, the brake fluid should be added with the decrease of oil level in the oilcan lest the oil inside is used up. The brake fluid should be added sufficiently when all air is exhausted.

制动踏板的自由行程由制动总泵推杆来调整。

The free travel of brake pedal is adjusted by the pushrod of brake master cylinder.

F、工作装置液压系统

F. Hydraulic system of working device

工作装置液压系统是由齿轮泵、多路换向阀、优先阀、举升臂油缸、翻斗油缸、转向油缸、工作油箱及油管等组成。压力油从齿轮泵经优先阀、多路换向阀，通过操纵相应的滑阀把油液输送到各油缸。工作装置不工作时，油液经多路换向阀的中间油道回油箱。

The hydraulic system of working device is made up by gear pump, multitandem direction valve, priority valve, lifting cylinder, tipping bucket cylinder, steering cylinder, operating oil tank and oil pipe. The pressure oil runs through the priority valve and multitandem direction valve from gear pump and oil fluid is transported to each oil cylinder by handling the corresponding slide valve. When working device is out of operation, the oil fluid runs back to the oil tank through the middle oil line of multitandem direction valve.

搅拌车电器设备

Electrical Equipment of Mixer Truck

自动上料搅拌车电器系统采用直流电压 24 伏，负极接地制。

The electrical system of automatic charging mixer truck uses 24 VDC with negative pole grounded.

一、蓄电池

I. Storage battery

蓄电池功用上是发动机起动时，供电给起动电动机；当发动机正常工作时，若发电机的电压低于蓄电池的电压时，则供电给用电设备；若发电机的电压高于蓄电池的电压时，则发电机向蓄电池充电，以储存一部分电能。

The storage battery is used to power the starting motor when the engine is started; when the engine functions properly, it will power the electric equipment if the voltage of generator is lower than that of storage battery; the generator will charge the storage battery to save part of electric power if the voltage of generator is higher than that of storage battery.

在使用中应注意：

Attention should be paid to the following items during use:

1、经常保持充电状态，如长期不用，每月最少应充电一次。

1. Storage battery should be charged regularly. If not used for a long time, it should be charged at least once every month.

2、经常检查电解液液面高度，液面应高出极板 10-15 毫米，不足时可添加蒸馏水。如确因电解液漏洒损失时，则应加比重为 1.28 的电解液。

2. The electrolyte level should be checked frequently, which should be higher than the plate by 10 - 15 mm. When the electrolyte is insufficient, distilled water may be added. If leakage, spill or loss of electrolyte is confirmed, electrolyte with proportion of 1.28 may be added.

3、应经常检查蓄电池通气孔是否堵塞，如若堵塞，应及时清除外部泥土和脏物，擦净溢出的电解液，并在电极上涂以薄层钙基润滑脂以防腐蚀。

3. Air hole of storage battery should be checked regularly to see whether it is blocked. Dirt or unclean matters should be cleaned and the overflowing electrolyte should be wiped out. In addition, thin layer calcium base grease should be applied on the electrode to prevent corrosion.

4、不得频繁使用起动机。

4. Starter should not be used frequently.

5、蓄电池应紧固，以免受震动而损坏。

5. Storage battery should be fastened well to prevent damage caused by vibration.

二、发电机与调节器

II. Generator and regulator

1、硅整流发电机：

1. Silicon rectification generator:

发电机上标有“+”、“F”两个接线柱和一个“—”接地钉。它们应与继电调节器上标有相同

文字的接线柱连接。

The generator has two wiring terminals marked as “+” and “F” and one “—” earthing screw, which should be connected to the wiring terminals marked with the same work on the relay regulator.

发电机两端用螺栓固定在发电机支架上；另有一个发电机撑脚，用来调整三角皮带张紧度。

Both sides of generator should be fastened on the generator support with bolts; there is one supporting lug of generator, which is used to adjust tensivity of triangle belt.

2、继电调节器

2. Relay regulator

继电调节器是由截流器、调压器、限流器三个装置所有组成。继电调节器上标有“+”、“F”二个接线柱和一个“—”接地螺钉，“+”、“F”接线柱应与发电机上相应的接线柱相连。

Relay regulator consists of cutoff device, voltage regulator and current limiter. The relay regulator has two wiring terminals marked as “+” and “F” and one “—” earthing screw. Wherein the “+” and “F” wiring terminals should be connected to the corresponding wiring terminals on the generator.

蓄电池的“+”接线柱与电流表“—”极相连，电流表的“+”极接线柱与电锁的一档相连。

The “+” wiring terminal of storage battery is connected to the “—” pole of ammeter, while the “+” of the ammeter should be connected to the first gear of electric shock.

发现调节器工作不正常或使用 1000 小时后，需要有熟练的电工进行调整。

In case it is found that the regulator works abnormally or has been working for 1,000 hours, a skillful electrician is needed to conduct adjustment.

三、起动机

III. Starter

它的用途是由蓄电池供给电源来起动发动机。起动机齿轮的转动是以电磁驱动，并带有单向滚柱式离合器，能防止起动机电枢随起动后的发动机高速旋转而损坏。

It is used to start the engine by using storage battery to supply power. Rotation of the starter gear is electromagnetically driven and one-way roller clutch is provided, which can prevent damage to the starter armature due to high-speed rotation along with the operating generator.

四、其他

IV. Others

1、电流表

1. Ammeter

电流表用以蓄电池的充、放电电流显示，当发动机向蓄电池充电时，指针偏向“+”；反之则为指向“-”极；无电流通过则为指零。

It is used to indicate charge and discharge current of the storage battery. When the generator charges the storage battery, the pointer will point to the “+”; conversely, it will point to the “-”, and if there is no current indication, the pointer will point to zero.

2、电锁

2. Electric lock

电锁控制整个电气系统，钥匙插入锁孔后，向右转一档时，所有电器的电源接通；向右转二档，起动机起动。为了保护启动马达，要求在发动机起动后，立即将电锁松开，钥匙自动复位至一档位置。

It is used to control the entire electrical system. After the key is inserted into the keyhole, all the electrical equipment will be connected to the power supply when you turn the key toward right side by one gear; if you turn the key toward right side by two gears, the starter will be started. In order to protect the starting motor, it is required to immediately release the electric shock after the engine is started, and the key will automatically reset to the first gear.

搅拌车油种类（见下表）

Oil category for mixer truck (see below)

油的牌号 Grade of Oil 加油部位 Position	
燃油箱 Fuel Tank	夏季（10℃以上）0 号，冬季（10℃以下）-10 号柴油 Summer (above 10℃): 0# diesel, winter (below 10℃): -10# diesel
油底壳 Oil Pan	夏季（10℃以上）CC40，冬季 CC30 柴油机润滑油 Summer (above 10℃): CC40#, winter: CC30 diesel lubricating oil
变速箱，驱动桥 Gearbox and drive axle	18#双曲线齿轮油 18# hypoid gear oil
液压油箱 Hydraulic Oil Tank	冬 46、夏 68 抗磨液压油（国际油） Winter: 46#, summer: 68# anti-wear hydraulic oil (international oil)
各注油嘴 Each Oil Injection Nozzle	组合钙基润滑油 Combined calcium-based lubricating oil
刹车油 Brake Oil	“912”合成制动液 “912” synthetic brake fluid

第六章 发生故障时的应急措施

Chapter VI Emergency Measures for Fault

故障及处理

Faults and Handling

6.1 搅拌罐不旋转

6.1 The mixing drum does not rotate

6.1.1 检查液压油油位和油品质

6.1.1 Check the oil level and quality of hydraulic oil

液压油不足时，补足液压油直至油位计中位

If the hydraulic oil is not sufficient, fill it until the middle position of oil level gauge

液压油很脏时，更换液压油

Replace hydraulic oil when it is very dirty

6.1.2 检查进油球阀是否被关闭

6.1.2 Check whether the inlet ball valve is closed or not

如关闭，开启球阀并用限位螺栓限位

If closed, open the ball valve and limit with stop bolt

6.1.3 检查滤清器是否已堵塞,若是则更换滤芯

6.1.3 Check whether the filter is blocked or not. If blocked, replace the filter element

6.1.4 检查控制连接结构及软轴

6.1.4 Check the control and connection structure and flexible shaft

将泵控制手柄向前或向后推动一段距离后，查看泵控制阀上控制柄是否相应的变更了角度。

After the pump control handle is pushed forward or back by certain distance, check whether the control handle on the pump control valve changes the angle accordingly.

如有相应变更角度，则转入 6.1.5 步骤

If it changes, jump to the Step 6.1.5

如未变更角度则检查如下步骤

If it does not, check in the following steps

①检查泵控制软轴是否与控制手柄有效连接

① Check whether the pump control flexible shaft is effectively connected to the control handle

②检查泵控制软轴是否固定牢固在软轴座上

② Check whether the pump control flexible shaft is firmly secured on the flexible shaft seat

③检查软轴座是否固定牢固在底架上，刚度是否足够

③ Check whether the flexible shaft seat is firmly secured on the chassis and stiff enough

④检查控制手柄变动角度，软轴是否伸缩。如不伸缩则软轴已断，更换软轴。

④ Check whether the control handle changes its angle and flexible shaft can extend or not. If the flexible shaft cannot extend, replace the flexible shaft.

⑤检查软轴是否与控制阀联接牢靠及软轴后端是否固定牢靠

⑤ Check whether the flexible shaft is firmly connected to the controller and backend of flexible shaft is secured firmly

⑥检查控制器是否已损坏，损坏则更换控制器

⑥ Check whether the controller is damaged. If damaged, replace the controller

注意：驾驶室内控制器与后控制器并联，如后控制器因配件损坏问题失效，则可用驾驶室控制器暂时代用，直至损坏配件更换。如果后控制器或驾驶室控制器皆因配件的损坏而失效，在当有混凝土在搅拌罐中时，司机可用液压泵上的控制阀的控制柄来强制卸料，注意使用此控制柄

只能慢慢的改变角度，且角度不宜过大，（大角度改变时，罐的转速将有可能很快，不安全）

Notes: the controller in the cab is connected to the rear controller in parallel. If the failure of rear controller occurs resulting from parts damaged, the controller in the cab can be used instead until the damaged parts are replaced. If the failure occurs in the rear controller or the controller in the cab resulting from the parts damaged, the driver can force discharge by using the control handle of control valve on the hydraulic pump when there is concrete being mixing in the mixing drum; however, the angle of control handle should only be changed slowly and should not be large (if it is changed in large angle, the rotating speed of drum may be too fast, which is not safe)

6.1.5 检查泵控制阀

6.1.5 Check the pump control valve

用手小角度拨动控制阀控制柄，查看罐体是否旋转

Check whether the drum rotates by manually turning the control handle of control valve in small angle 如旋转，则转入 6.1.6 步骤

If it rotates, jump to the Step 6.1.6

如不旋转，则检查如下步骤

If it does not, check in the following steps

①检查控制柄与阀旋转轴是否连接牢固和一并旋转

① Check whether the control handle and rotation axle of valve are firmly connected and rotate together

②从泵上拆下控制阀总成时，检查控制阀的连杆机构是否与伺服机构的连杆联接，如未联结，则联接起来，如连杆损坏，则更换连杆

② When the control valve assembly is disassembled from the pump, check whether the connecting rod mechanism of control valve is connected to that of servo mechanism; if they are not connected, connect them, while if the connecting rod is damaged, replace the connecting rod

③查看控制阀上的 3 个控制油孔是否干净，如有堵塞，需用气枪冲洗干净。同时也应检查泵体上进入伺服机构的控制油孔是否堵塞。

③ Check whether 3 control oil holes on the control valve are clean; if they are blocked, rinse thoroughly with air gun. Meanwhile, check whether the oil hole on the pump to the servo mechanism is blocked.

④拨动控制柄，查看阀芯是否运动正常。如拨动控制柄，阀芯不运动，则阀芯断裂，更换控制器总成

④ Check whether the valve element runs normally by turning the control handle. If the valve does not run by turning the control handle, the valve element fractures, then the controller assembly should be replaced

6.1.6 检查泵补油压力

6.1.6 Check the oil refill pressure of pump

补油压力正常（18bar），则检查高压安全阀，如高压不足，则更换高压安全阀。

If the oil refill pressure is normal (18 bar), check the high-pressure safety valve; if the high pressure is not sufficient, replace the high-pressure safety valve.

补油压力为 0，则泵输入轴或花键损坏，更换泵输入轴

If the oil refill pressure is 0, the input shaft of pump or spline is damaged, the input shaft of pump should be replaced

补油压力不正常—检查滤清器

If the oil refill pressure is not normal - check the filter

6.2 搅拌罐只能向一个方向旋转

6.2 The mixing drum can only rotate unidirectionally

6.2.1 检查控制器和控制软轴

6.2.1 Check the controller and control flexible shaft

将控制手柄向不旋转的方向推到最大行程。查看控制阀上控制柄是否相应变动控制角度

By pushing the control handle towards the direction of non-rotation to the maximum travel, check whether the control handle on the control valve changes the angle accordingly

相应角度变化，但罐体不旋转，则转入 6.2.2 步骤

If it changes, the drum does not rotate, jump to the Step 6.2.2

无相应角度变化，则检查如下步骤：

If it does not, check in the following steps

测量下当前软轴行程尺寸，然后将控制手柄推到中停位置，再次记下当前软轴的行程尺寸。将控制软轴与控制阀控制柄相连接的万向球头脱离，然后将控制手柄向旋转方向推动到最大行程处，记录下当前行程尺寸。以中停位置为基准 0 行程，并确认最大不转方向和最大旋转方向，软轴的行程是否伸（缩）向相反行程方向。

Measure the travel size of current flexible shaft and push the control handle to the middle stop position, then note the travel size of current flexible shaft again. Disconnect the universal ball joint between the control flexible shaft and control handle of control valve, then push the control handle towards the direction of rotation to the maximum travel and note the current travel size. When the travel is 0 on the basis of middle stop position, confirm the maximum direction of non-rotation and maximum direction of rotation and whether the travel of flexible shaft extends (retracts) towards the reverse direction.

如以中停为基准 0 行程，软轴有相反行程方向，则该软轴已损坏，更换软轴。

If the travel is 0 on the basis of middle stop position, the flexible shaft travels reversely, which indicates that the flexible shaft is damaged, then the flexible shaft needs to be replaced.

如以中停为基准 0 行程，软轴无相反行程方向，则拆开控制器，查看控制器是否损坏，如无损坏，因控制器行程出现单向，则需要重新调整控制器行程，直至以中停为参照，向不同方向推动时，软轴伸缩双向行程尺寸相近，调整好后再次联接万向球头。

If the travel is 0 on the basis of middle stop position, the flexible shaft does not travel reversely, disassemble the controller and check whether the controller is damaged; if it is not damaged, the travel of controller needs to be readjusted as the controller travels unidirectionally so that the controller is pushed towards different directions on the basis of middle stop position, the travel sizes of extension and retraction of flexible shaft are close. After that, reconnect the universal ball joint.

6.2.2 检查控制阀内外联接（脱离控制软轴与控制柄相联的万向球头）

6.2.2 Check the internal and external connection of control valve (disconnect the universal ball joint between control flexible shaft and control handle)

a>检查控制阀输入轴是否磨损而导致单边失效

a> Check whether the unilateral failure occurs in the input shaft of control valve due to wear

b>将控制阀总成拆卸下，检查内部连杆机构是否损坏

b> Disassemble the control valve assembly and check whether the internal connecting rod mechanism is damaged

C>检查控制阀上控制油孔是否有因液压油脏而堵塞,如油脏堵塞控制油孔需用压力气枪清洗干净。

c> Check whether the control oil hole on the control valve is blocked resulting from dirty hydraulic oil. If it is, the control oil hole needs to be rinsed thoroughly with high-pressure air gun.

d>检查泵主体壳上到伺服机构的控制油孔是否因脏油而堵塞，如被堵塞，用压力气枪清洗干净。

d> Check whether the control oil hole on the pump body shell to the servo mechanism is blocked resulting from dirty hydraulic oil. If it is, the control oil hole needs to be rinsed thoroughly with high-pressure air gun.

6.2.3 检查补油单向阀

6.2.3 Check the one-way valve of oil refill

检查补油单向阀,确认是否被堵塞或损坏,如堵塞需拆下在干净的液压油中清洗,如损坏则更换。

Check the one-way valve of oil refill and confirm whether it is blocked or damaged. If it is blocked, disassemble and wash it in the clean hydraulic oil; while if it is damaged, replace it.

6.2.4 检查不旋转方向相对应的安全阀及平衡梭阀

6.2.4 Check the safety valve and balance shuttle valve corresponding to the safety valve and balance shuttle valve

测试不旋转方向相应的高压压力,如无高压则拆下相应的安全阀检查是否被堵塞或损坏,如被堵塞,则清洗,如损坏则更换。

Test the pressure of high pressure corresponding to the direction of non-rotation. If there is no high pressure, disassemble the corresponding safety valve and check whether it is blocked or damaged. Clean in case of block and replace in case of damage.

检查梭阀,查看相应方向弹簧是否损坏,而导致单边直接泄压,更换弹簧或增加垫块。

Check the shuttle valve and view whether the spring of corresponding direction is damaged, resulting in direct pressure relief unilaterally. If it is, replace the spring and add pad.

注:6.2.3,6.2.4 两步骤只能是受过初步液压系统专业维修训练的技工方能进行。

Notes: the Step 6.2.3 and Step 6.2.4 are only be implemented by the technician who has received the professional maintenance training relating to hydraulic system.

6.3 系统反应迟缓

6.3 Slow response of system

6.3.1 检查取力器转速(可由电发动机转速折算)至少 500-600rpm

6.3.1 Check the rotating speed (can be converted from the rotating speed of electric engine) of power takeoff with 500 - 600 rpm at least

6.3.2 检查液压油油位及油品质

6.3.2 Check the oil level and quality of hydraulic oil

液压油不足, 补充液压油直至油位计中位

If the hydraulic oil is not sufficient, fill it until the middle position of oil level gauge

液压油很脏,更换液压油

Replace hydraulic oil when it is very dirty

6.3.3 检查滤清器

6.3.3 Check the filter

如滤清器真空表指针指向或超过-2.5psi(1.27cm of Hg)则更换滤芯

If the pointer of vacuum gauge of filter points to or exceeds -2.5 psi (1.27 cm of Hg), replace the filter element

6.3.4 检查液压泵补油压力

6.3.4 Check the oil refill pressure of hydraulic pump

6.3.5 检查马达的回油被压阀

6.3.5 Check the oil return back pressure valve

6.3.6 检查泵控制阀内外联接件

6.3.6 Check the internal and external connecting parts of pump control valve

6.4 系统油温过高

6.4 Oil temperature of system is too high

6.4.1 检查液压油油位及油品质

6.4.1 Check the oil level and quality of hydraulic oil

液压油不足时, 补足液压油直至油位计中位

If the hydraulic oil is not sufficient, fill it until the middle position of oil level gauge

液压油很脏, 则更换液压油

Replace hydraulic oil when it is very dirty

6.4.2 检查进油球阀是否被关闭

6.4.2 Check whether the inlet ball valve is closed or not

如被关闭, 开启球阀并用限位螺栓限位

If closed, open the ball valve and limit with stop bolt

6.4.3 检查滤清器是否堵塞

6.4.3 Check whether the filter is blocked

查看真空表, 如表针指向或超过 -2.5psi (1.27cm of Hg) 则更换滤芯

If the pointer of vacuum gauge of filter points to or exceeds -2.5 psi (1.27 cm of Hg), replace the filter element

6.4.4 查看散热风扇是否旋转

6.4.4 Check whether the cooling fan rotates or not

不旋转进入 6.4.5 步骤

If it does not rotate, jump to the Step 6.4.5

如旋转则检查 6.4.6 步骤

If it rotates, check items provided in the Step 6.4.6

6.4.5 检查散热器温控开关及电路

6.4.5 Check the temperature-controlled switch and circuit of radiator

开启手动开关, 查看风扇是否旋转。

Check whether the fan rotates or not by opening the manual switch.

<A> 不旋转, 则检查

<A> Check the following items if it does not rotate

a> 保险丝是否被熔断, 熔断则更换保险丝 (15A)

a> Check whether fuse is blown. Replace the fuse (15A) if it is blown

b> 检查线路是否接触不良, 线路是否被扯断(接触不良, 则重新连接, 线路扯断则接通)

b> Check whether the line is poor contact or pulled apart (reconnect in case of poor contact and connect in case of pull-apart)

c> 检查开关是否损坏, 损坏则更换开关

c> Check whether the switch is damaged. If damaged, replace it

 旋转, 则关闭手动开关检查

 Close the switch manually and check the following items if it rotates

a> 连通温控开关的线路是否接触不良或线路被扯断(接触不良, 则重新连接, 线路扯断则接通)

a> Check whether the line connecting the temperature-controlled switch is poor contact or pulled apart (reconnect in case of poor contact and connect in case of pull-apart)

b> 将温控开关触头置于 60℃ 以上的热水中, 查看散热器风扇是否旋转, 如不旋转, 更换稳控开关。

b> Check whether the fan of radiator rotates by putting the contact of temperature-controlled switch in the water of above 60°C. If it does not rotate, replace the temperature-controlled switch.

注意: 因手动开关和温控开关并联控制风扇, 如温控开关损坏, 可用于手动开关暂时手动控制风扇直至更换温控开关

Notes: As the fan is controlled by the manual switch and temperature-controlled switch connected in parallel, the fan can be controlled temporarily with manual switch until the temperature-controlled switch is replaced if the temperature-controlled switch is broken.

6.4.6 检查补油压力和马达的壳体压力

6.4.6 Check the oil refill pressure and case pressure of motor

用压力表测试该项压力,标准压力为 18bar,如果压力过低更换补油泵或马达壳体背压阀。

Test the pressure with pressure gauge. The standard pressure is 18 bar, and replace the oil refill pump or back pressure valve of motor case if the pressure is too low.

6.5、紧急处理预案

6.5. Emergency Scenario

当搅拌车在运输混凝土时,如果出现发动机或液压泵或马达或齿轮箱损坏导致搅拌罐无法旋转,为了避免混凝土凝固在罐体内,请用如下紧急处理预案:

When the mixer truck is transporting the concrete, if the mixing drum cannot mix as the engine, hydraulic pump, motor or gearbox is damaged, the following emergency scenarios should be implemented to prevent the solidification of concrete in mixing drum:

(1) 发动机损坏

(1) Engine is damaged

因发动机损坏,搅拌车将失去动力源,故罐体无法旋转,请采用如下紧急处理预案:

If the engine is damaged, the mixing drum will not rotate as the mixer truck loses power, then the following emergency scenario should be implemented:

预案 A: 从维修孔处将混凝土排出

Scenario A: discharge the concrete from maintenance hole

首先将罐体后锥体上的 2 个维修孔盖的螺栓拆除,并将维修孔盖打开,再将液压泵与马达相连的两根高压管旋转松或拆除,将混凝土从维修孔处排出。因维修孔在罐体的下端,故混凝土从入孔流出的速度将很快,在打开维修孔前,尽量使维修孔朝向车侧面。

First remove the bolts of 2 maintenance hole covers on the rear cone of drum body and open the maintenance hole covers, then loosen or remove two high-pressure pipes between the hydraulic pressure pump and motor, finally discharge the concrete from maintenance hole. As the maintenance hole is located at the bottom of drum body, the concrete will run out of the hole quickly. Before opening the maintenance hole, make the maintenance hole towards the side of truck as far as possible.

注意: 在旋松或拆除高压管时,因系统中有高压阻力,在旋松管件时有可能会有高压油喷出,请特别注意安全并防止烫伤。另罐体由于惯性作用,有可能自动旋转,救援人员必须注意安全。在旋松或拆除高压油管前,在滚轮与轨道之间放入木楔,阻碍罐体旋转。

Notes: When the high-pressure pipe is loosened or removed, the high pressure oil may inject as there is high-pressure resistance in the system, so please be careful and prevent burns. In addition, the drum body may rotate automatically due to the inertia effect, so the rescuers must pay attention to safety. Before loosening or removing the high-pressure pipes, put wooden wedge between the roller and track to prevent rotation of drum body.

预案 B: 紧急救援

Scenario B: Emergency rescue

①准备工作

① Preparations

准备好一套救援软管(2 根高压软管,1 根低压软管),并准备 18L 液压油。

Prepare a set of rescue hoses (including 2 high-pressure hoses and 1 low-pressure hose) and 18 L hydraulic oil.

②故障车的操作

② Operations of faulty truck

- a、将故障车熄火，并用手刹制动。（如在坡度处，应用于木楔限位后轮）
- a. Stall the faulty truck and brake it with handbrake. (The wooden wedge should be used to stop the rear wheels in the slope)
- b、将高压管和低压管从马达接口处卸下。注意用容器将泄出的油盛起来，避免污染车辆。
- b. Remove the high-pressure hose and low-pressure hose from the interface of motor. Contain the drained oil with container to prevent it from polluting the truck.
- c、将 3 根救援管分别接到相应的接口位置，必须紧固各接口。
- c. Connect 3 rescue hoses to the corresponding interfaces and tighten each interface.
- d、将卸料槽放低并固定，在卸料槽尾端准备好盛放混凝土的料斗或容器。
- d. Lower and secure the discharge chute and get the hopper or container ready at the end of discharge chute to contain the concrete.

③救援车的操作（未装载车辆）。

③ Operations of rescue vehicle (unloaded vehicle).

- a、将同型号液压系统的搅拌车驾驶到事故现场，将救援车与故障车并排放置，平行间距约 1-2 米，车头与车头平齐。
- a. Drive the mixer truck with same model of hydraulic system to the site of accident and place the faulty vehicle and rescue vehicle side by side with parallel distance of about 1 to 2 meters and abreast vehicle heads.
- b、将发动机熄灭，并用手刹制动。（如在坡度处，应用于木楔限位后轮）
- b. Shut down the engine and brake with handbrake. (The wooden wedge should be used to stop the rear wheels in the slope)
- c、将高压管和低压管从马达接口处卸下。（注意用容器将泄出的油盛起来，避免污染车辆）。
- c. Remove the high-pressure hose and low-pressure hose from the joint of motor. (Contain the drained oil with container to prevent it from polluting the truck).
- d、将 3 根救援管参照故障车的联接位置安装到救援车液压泵相应的接口位置，必须紧固各接口。
- d. Install 3 rescue hoses on the corresponding interfaces of hydraulic pump of rescue vehicle by referring to the connection positions of faulty vehicle and tighten each interface.
- e、启动救援车的发动机，使发动机怠速运转数分钟后，在救援车液压油箱中补足液压油。
- e. Start the engine of rescue vehicle and keep it running for several minutes in idle speed and replenish the hydraulic oil tank of rescue vehicle with hydraulic oil.
- f、操作救援车的控制系统，将故障车内混凝土排出。
- f. Operate the control system of rescue vehicle and discharge the concrete out of the faulty vehicle.
- g、救援完毕后，使两车恢复原状。
- g. Restore two vehicles to original state after the rescue is completed.

注意：救援软管需另外订购。2 根高压软管总成，1 根低压软管总成，皆长约 4.5 米。

Notes: the rescue hoses are required to be purchased separately. 2 high-pressure hoses assembly and 1 low-pressure hose assembly are about 4.5 meters in length.

（2）液压泵损坏

(2) The hydraulic pump is damaged.

预案 A：更换同型号的液压泵，使搅拌车恢复正常运转，将混凝土运到工地，再卸出。

Scenario A: Replace it with the hydraulic pump with same model and restore the mixer truck to normal operation to transport the concrete to construction site and discharge.

预案 B：从维修孔将混凝土排出，见发动机损坏预案 A。

Scenario B: Discharge the concrete from maintenance hole, please see the Scenario A for “The engine is damaged”.

预案 C：紧急救援，见发动机损坏预案 B。

Scenario C: Emergency rescue, please see the Scenario B for “The engine is damaged”.

(3) 马达损坏

(3) The motor is damaged

预案 A: 更换同型号马达, 使搅拌车恢复正常运转, 将混凝土运到工地再卸出。

Scenario A: Replace it with the motor with same model and restore the mixer truck to normal operation to transport the concrete to construction site and discharge.

预案 B: 从维修孔将混凝土排出, 见发送机损坏预案 A。

Scenario B: Discharge the concrete from maintenance hole, please see the Scenario A for “The engine is damaged”.

(4) 齿轮箱的损坏

(4) The gearbox is damaged

预案 A: 向罐体投入大量的缓凝剂和水或葡萄糖溶剂, 缓解混凝土凝固, 运到工地再处理

Scenario A: Input a large amount of retarder and water or glucose solvent into the drum to delay the solidification and handle it on the construction site

预案 B: 从维修孔将混凝土排出, 见发动机损坏预案 A。

Scenario B: Discharge the concrete from maintenance hole, please see the Scenario A for “The engine is damaged”.

一般故障及原因分析

General Faults and Cause Analysis

名称 Name	故障现象 Faults	原因分析 Cause analysis	排除方法 Troubleshooting
一、发动机 I. Engine		请参看柴油机说明书 Please refer to Instructions of Diesel Engine	
二、离合器 II. Clutch	1. 接合时打滑 1. Clutch slips when engaging	1. 踏板自由行程过小 1. Free travel of pedal is too short 2. 压力弹簧软弱 2. Pressure spring is springless 3. 磨擦片表面有油污 3. Friction plate is polluted by oil 4. 磨擦片磨损逾期 4. The friction plate is worn and exceeds the service life	重新调整 Readjust 更换压力弹簧 Replace the pressure spring 清洗 Clean 更换磨擦片 Replace the friction plate
	2. 接合时发抖 2. The loader trembles when clutch is engaging	1. 花键磨损过大 1. Spline is excessively worn 2. 各紧固螺栓松动 2. Each fastening bolt is loose 3. 磨擦面有油脂 3. Friction surface is polluted by oil 4. 分离杠杆调整不均 4. Disengaging lever is unevenly adjusted	更换离合器轴或轴 Replace clutch shaft or shaft 紧固 Fasten 清洗 Clean 重新调整 Readjust

	3. 离合器不易分离 3. The clutch is hard to disengage	1. 踏板自由行程过大 1. Free travel of pedal is too long 2. 分离杠杆调整不当 2. Disengaging lever is improperly adjusted	重新调整 Readjust 重新调整 Readjust
III. Gearbox 三、变速箱	1. 发响声 1. Produce noise	1. 齿轮磨损逾期、齿的侧隙过大 1. The gear is worn and exceeds the service life and backlash is too large 2. 轴承磨损逾期 2. The bearing is worn and exceeds the service life 3. 各紧固螺栓松动 3. Each fastening bolt is loose 4. 润滑油不足 4. The lubricating oil is insufficient 5. 齿轮与轴的花键过分磨损 5. The gear and shaft splines are excessively worn	更换齿轮 Replace the gear 更换齿轮 Replace the gear 紧固 Fasten 补充润滑油 Replenish the lubricating oil 更换齿轮或轴 Replace the gear or shaft
	2. 时常跳档 2. The gear lever often jumps back to the neutral position	1. 拨叉轴定位弹簧软弱或失效 1. The guide spring of shift rail is springless or fails 2. 拨叉轴定位槽磨损超限 2. The guide slot of shift rail is worn and exceeds the service life 3. 内外花键磨损 3. The internal and external splines are worn	更换弹簧 Replace the spring 更换拨叉轴 Replace the shift rail 更换齿轮或轴 Replace the gear or shaft
	3. 换档不灵活 3. The gear shifting is not flexible	1. 齿轮齿端碰毛 1. The end of gear touches burr	修去毛刺或更换齿轮 Deburr or replace the gear
IV. Drive axle 四、驱动桥	1. 行驶时有响声 1. Produce noise during driving	1. 主减速齿轮啮合点不好 1. The gear engaging point of the main reduction gear is not good 2. 轴承磨损超限或松动 2. The bearing is excessively worn or loose 3. 齿轮磨损超限 3. The gear is worn and exceeds the service life	调整或更改锥齿轮 Adjust or replace the bevel gear 更换轴承或调整 Replace or adjust the bearing 更换齿轮 Replace the gear
	2. 制动时发响 2. Produce noise during braking	1. 制动底板弯曲 1. The braking plate is crooked 2. 制动摩擦衬片铆钉松动 2. The rivet of braking friction liner is loose	修复或更换 Repair or replace 修理 Repair 修理或更换

		3. 制动鼓损坏 3. The brake drum is damaged	Repair or replace
	3. 制动时车子跑偏 3. Deviation occurs during braking	1. 制动蹄片表面有油 1. The brake shoes are polluted by oil 2. 间隙调整不当 2. The clearance is adjusted improperly 3. 轮胎气压不合标准 3. The tire pressure is not up to standard	清洗 Clean 重新调整 Readjust 使气压一致 Keep the same pressure
	4. 制动不灵 4. Ineffective braking	1. 制动鼓与制动蹄片间隙调整不当 1. The clearance between brake drum and brake shoes is adjusted improperly 2. 有油污 2. There is oil stain 3. 制动磨擦衬片磨损超限 3. The braking friction liner is worn and exceeds the service life	重新调整 Readjust 清洗 Clean 更换 Replace
V. Steering system 五、转向系统	1. 方向盘慢转轻、快转沉 1. The slow turning of steering wheel is light, while the fast turning is heavy	供油不足 Insufficient oil supply	调整分流阀、优先阀 Adjust the flow divider and priority valve
	2. 转身无力 2. Unable to turn	工作压力油压力低 Low working pressure and oil pressure	调整优先阀溢流阀 Adjust the priority valve and overflow valve
	3. 转动方向盘油缸不动 3. The oil cylinder does not move when turning the steering wheel	系统中有空气或油量不足 The air exists in the system or oil is insufficient	排除空气或补充油 Remove the air or supplement oil
VI. Hydraulic System & Braking System 六、液压系统、制动系统	1. 举升臂提升力不足或转斗力不足 1. Insufficient lifting force or rotating force of lifting arm	1. 安全阀调整不当系统压力低 1. Low system pressure resulting from the improper adjustment of safety valve 2. 吸油管及滤油器堵塞 2. The oil suction pipe and oil filter are blocked 3. 油泵、油缸、管路内漏 3. Internal leakage of the oil pump, oil cylinder and pipeline 4. 多路阀过度磨损、阀杆与阀体配合间隙超过规定 4. The multitandem valve is excessively worn and the clearance between connecting rod and valve body exceeds the specified	按规定值调整系统工作压力 Adjust the working pressure of system based on specified value 清洗换油 Clean and change oil 更换油泵、并按自然沉降量检查系统密封性 Replace the oil pump and check the air-tightness of system based on the natural precipitates 更换多路阀 Replace the multitandem valve

2. 系统工作性能降低或不稳定 2. Low or unstable working performance of system	1. 工作油变质 1. Deterioration of operating oil 2. 异物堵塞管路 2. The pipeline is blocked by foreign matters 3. 滤油器堵塞或损坏 3. The filter is blocked or damaged 4. 系统内有空气 4. There is air in the system	更换工作油 Replace the operating oil 清洗油路系统和油箱 Clean the oil line system and oil tank 清洗或更换 Clean or replace 检查进油系统有无漏气 Check whether the oil inlet system leaks or not
3. 举升臂举升后自行下沉 3. The lifting arm lowers automatically after being lifted	1. 举升臂油缸内漏 1. Internal leakage of oil cylinder of lifting arm 2. 多路阀阀杆间隙过大 2. Clearance between valve rods of multitandem valve is too large	拆修油缸、更换密封圈 Disassemble and repair the oil cylinder and replace the seal ring 拆修或更换 Repair or replace
4. 油温过高 4. Oil temperature too high	1. 生负荷工作时间过长 1. Excessive working hours at full-load 2. 油量不足 2. Oil is insufficient	停机休息或减少负荷 Shutdown and reduce the load 加油到规定油位 Add oil to the specified oil level
5. 方向盘回位后继续转向 5. Steering wheel continues turning after return	1. 转向器内回位弹簧片损坏 1. Return spring leaf in steering gear is damaged 2. 配油套和配油轴之间卡死或配油套与阀体之间卡死 2. The oil distribution sleeve and shaft are jammed or the oil distribution shaft and valve body are jammed	拆修更换弹簧片 Repair or replace the spring leaf 拆开转向器修复 Disassemble the steering gear and repair
6. 脚制动力不足 6. Foot brake force is insufficient	1. 制动分泵漏油 1. The brake cylinder has an oil leakage 2. 制动液压管路中有空气 2. Air exists in the brake hydraulic line 3. 制动总泵皮碗损坏 3. The leather cup of brake master bump is damaged 4. 制动总泵油液不足 4. The oil of brake master bump is insufficient 5. 推杆行程调节不当 5. The travel of push rod is improperly adjusted 6. 制动磨擦片磨损到极限	更换分泵油封 Replace the oil seal of pump 排除空气 Remove the air 更换皮碗 Replace the leather cup 加油 Add oil 调正行程 Adjust the travel 更换新制动磨擦片 Replace it with new braking friction plate

		6. The braking friction plate is fully worn	
七、 电 气 系 统 VII. Electric system	1. 发动机正常而蓄电池不充电或充电率低 1. The engine is normal, while the storage battery is not charged or its rate of charge is low	1. 蓄电池极板硫化 1. The plate of storage battery is vulcanizing 2. 发电机皮带过松或损坏 2. The generator belt is too loose or damaged 3. 接线不牢, 接触不良 3. The connection is not firm, and the contact is poor 4. 调节器调节不当或有损坏 4. The regulator is improperly adjusted or damaged	脱硫处理或更换极板 Desulphurize or replace the plate 重新调整或更换 Readjust or replace 检查并清除之 Check and clear it 重新调整或修理 Readjust or repair
	2. 蓄电池容量不足 2. Capacity of storage battery is insufficient	1. 电解液比重或液面过低 1. The specific gravity or fluid level of electrolyte is too low 2. 极板间短路 2. There is short circuit in plates 3. 极板硫化 3. The plate is vulcanizing 4. 导线接触不良 4. The wire is in poor contact 5. 极板活性物质脱落 5. The plate active material is falling off	重新调整比重或添加电解液 Readjust the specific gravity or add electrolyte 消除沉淀物, 更换电解液 Remove the precipitate and replace the electrolyte 脱硫处理或更换极板 Desulphurize or replace the plate 检查并消除之 Check and remove it 更换极板 Replace the plate
	3. 发电机不发电 3. The generator does not generate electricity	1. 剩磁消失 1. The residual magnetism vanishes 2. 磁场线圈回路断 2. The field coil is in open circuit 3. 整流子接触不良 3. The commutator is in poor contact 4. 电刷卡住不灵活 4. The electric brush shifting is stuck and not flexible 5. 电枢匝间短路 5. There is short circuit in armature turn-to-turn	1、按发电机原来极性, 用蓄电池 1. Use the storage battery according to the original polarity of the generator 2、接于磁场线圈两端并接通 2. Attach it to the ends of the field coil and make it connected 3、用 0 号或 00 号砂纸磨光 3. Use #0 or #00 abrasive paper to polish 4、修正电刷尺寸, 调整弹簧

			压力 4. Correct the size of electrical brush and adjust the spring pressure 5、检查并修复 5. Check and repair
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保修期限

Warranty Period

内容 Description	型号 Model	保修期限（月） Warranty Period (month)
搅拌罐罐体 Body of mixing drum	各种型号 Various models	6
底架 Chassis	各种型号 Various models	6
进料斗 Charge hopper		6
卸料槽 Discharge chute		6
延伸料槽 Extended chute		6
液压泵 Hydraulic pump		6
液压马达 Hydraulic motor		6
齿轮箱 Gearbox		6
散热器 Radiator	风冷 Air cooling	6
传动轴 Transmission shaft		6
减压阀 Reducing valve	7 KG	6
稳压阀 Pressure maintaining valve	2 KG	6
单向阀 One-way valve		6
卸压阀 Relief valve	3 KG	6
控制器 Controller	WJ116, WJ105, WJ101	6
控制软轴 Control flexible shaft	33C, 44C	6
轨道保护带 Track guard belt		6

用户须知

Instructions for Users

一、用户在购机之时，要领取说明书，本机合格证、保修卡

I. User should obtain the instructions, product certificate and warranty card at the time of purchase

二、驾驶员在操作本机前，必须详细阅读使用说明书，应按说明书中的要求进行操作、调试和保养。

II. Before operation, the driver must read the Operation Manual carefully and perform the operations, debugging and maintenance as required in the instructions.

三、本机的液压系统不准随意调整压力

III. The pressure of hydraulic system shall not be adjusted at will

四、保修期从购机之日起六个月。无产品合格证、保修卡、电器件、易损件及因使用不当造成的故障不在保修范围之内。轮胎在保修期内出现起包现象给予三包，其它情况不在保修范围之内。因缺黄油造成的损失不予三包。

IV. The warranty shall remain effective for six months from the date of purchase. The electric parts, wearing parts and faults resulting from misuse are not covered in the scope of warranty in case the product certificate and warranty card are not available. Except the swelling of tire, other cases are not covered in the Three-Guarantee during warranty period. The losses resulting from lack of grease are not covered in the Three-Guarantee.

五、本机出现机械故障时，用户应马上把本机型号，出厂编号，购买日期按保修卡的电话通知经销商。

V. When the fault occurs in the product, the user should notify the dealer of the model, factory number and date of purchase through the telephone provided in the warranty card.